abstract

Eqaab (عقاب) is an intelligent autonomous aerial surveillance system designed to overcome the limitations of fixed cameras and ground patrols in wide-area security zones. Built on a PX4-controlled quadrotor with a Raspberry Pi 5 companion computer, the system runs a dual-model YOLOv8 detection pipeline entirely onboard — identifying persons, vehicles, and hostile drones in real time without cloud dependency. Detected targets are tracked via DeepSORT, classified through an IFF node, and transmitted as structured GPS-tagged alerts over 4G LTE to a FastAPI cloud backend and a live React.js ground control dashboard. Eqaab delivers a complete end-to-end aerial security platform — from autonomous flight to real-time operator awareness — built and validated by a seven-member Computer Engineering team at Taif University.

Introduction

Modern security operations face growing challenges in monitoring wide-area zones such as national borders, critical infrastructure, and restricted facilities. Traditional surveillance systems — fixed cameras and ground patrols — suffer from limited coverage, blind spots, and heavy reliance on human operators prone to fatigue and delayed response. The integration of autonomous UAVs with artificial intelligence offers a transformative solution: a mobile, intelligent aerial platform capable of continuous surveillance, real-time threat detection, and instant operator notification.

Problem Statement

Large high-security areas continue to rely on surveillance infrastructure that is fundamentally inadequate for the scale and complexity of modern threats. Fixed cameras and ground sensors fail to provide full coverage across wide or difficult terrains, leaving dangerous blind spots where vehicles and intruders can move undetected. Excessive dependence on human operators introduces fatigue and cognitive overload, causing critical events to be missed or recognised too late for an effective response. When a security breach does occur, traditional systems lack the ability to continuously track the target or log its precise location — making incident review and lessons learned nearly impossible. These compounding limitations — restricted coverage, human error, slow response, and poor documentation — represent a critical gap that Eqaab is designed to close.

Objectives

The Eqaab system was developed to achieve the following objectives:

1. Design and implement an autonomous drone-based surveillance platform capable of executing full missions — from takeoff to landing — without continuous operator input.
2. Deploy a real-time AI detection pipeline onboard the drone to identify and classify persons, vehicles, and hostile drones using a dual-model YOLOv8 architecture.
3. Implement an IFF (Identification Friend or Foe) classification system to distinguish authorised friendly aircraft from unknown aerial targets before alert generation.
4. Fuse all detections with real-time GPS coordinates to produce georeferenced alerts transmitted instantly to a cloud backend over a 4G LTE link.
5. Develop a live React.js ground control dashboard providing operators with real-time situational awareness, detection alerts, and high-level mission command capability.

Methodology

The Eqaab system was developed following a structured engineering methodology divided into four sequential phases:

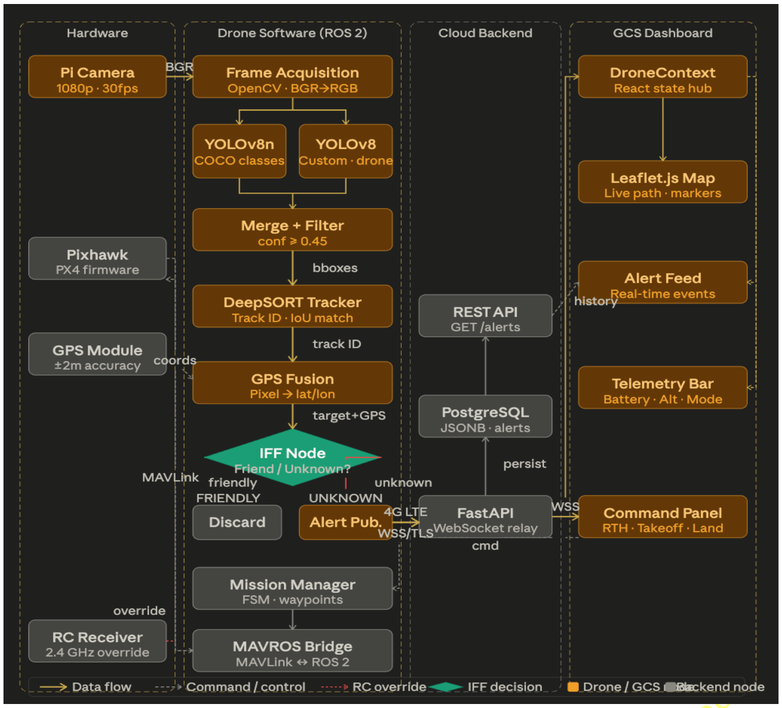
Phase 1 — Research and Design: The team conducted a comprehensive review of related UAV surveillance systems and AI-based detection frameworks. System requirements were defined, the three-tier architecture was designed, and hardware components were selected based on computational capacity, weight constraints, and operational range requirements.

Phase 2 — Hardware Integration: The drone platform was assembled and calibrated, integrating the Pixhawk flight controller, Raspberry Pi 5 companion computer, Pi Camera, GPS module, and 4G LTE dongle into a single cohesive airframe. PX4 firmware was configured and failsafe parameters were validated using QGroundControl.

Phase 3 — Software Development: The full software stack was implemented in parallel across three layers — the ROS 2 onboard node architecture, the FastAPI cloud backend, and the React.js GCS dashboard. The dual-model YOLOv8 detection pipeline was trained on Kaggle using a merged Roboflow dataset and deployed on the Raspberry Pi 5. The IFF classification node and DeepSORT tracking integration were implemented and validated independently before full-stack integration.

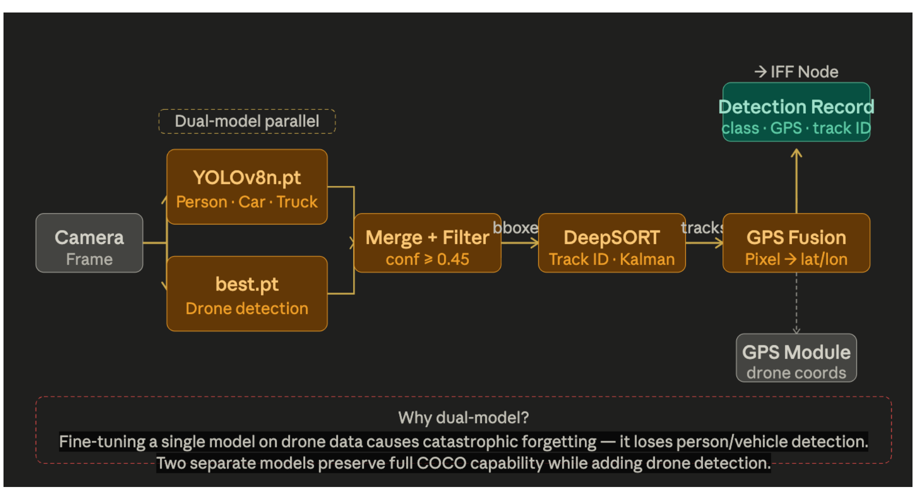
Phase 4 — Testing and Validation: The autonomous flight logic was validated in a Gazebo simulation environment with ArduPilot SITL before physical deployment. Detection performance was evaluated across all three target classes. End-to-end system latency, communication reliability, and GCS dashboard responsiveness were measured under realistic operating conditions.

System Architecture



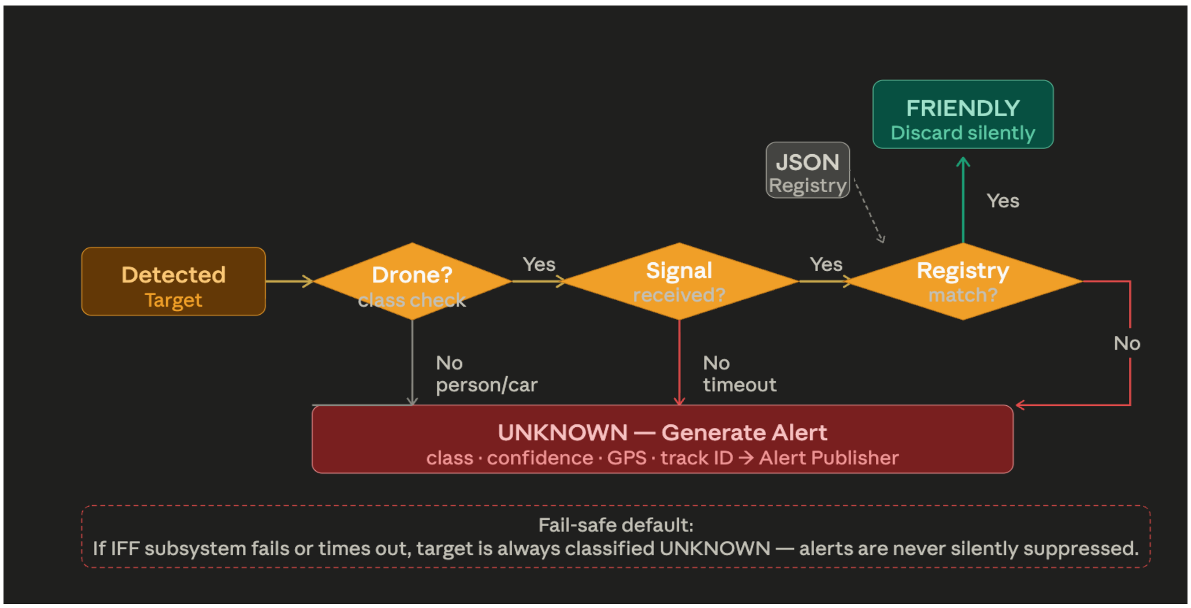
AI Detection Pipeline

The Eqaab detection pipeline runs entirely on the Raspberry Pi 5 without GPU acceleration. A dual-model architecture was adopted to overcome the catastrophic forgetting problem — fine-tuning a single YOLOv8 model on drone detection data caused it to lose the ability to detect persons and vehicles. The solution runs two independent models in parallel: `yolov8n.pt`, the official COCO pre-trained model detecting persons and vehicles, and `best.pt`, a custom-trained model detecting drones. Their outputs are merged, filtered at a confidence threshold of 0.45, and passed to DeepSORT for multi-object tracking. Each confirmed track is fused with real-time GPS coordinates to produce a georeferenced detection record forwarded to the IFF node.



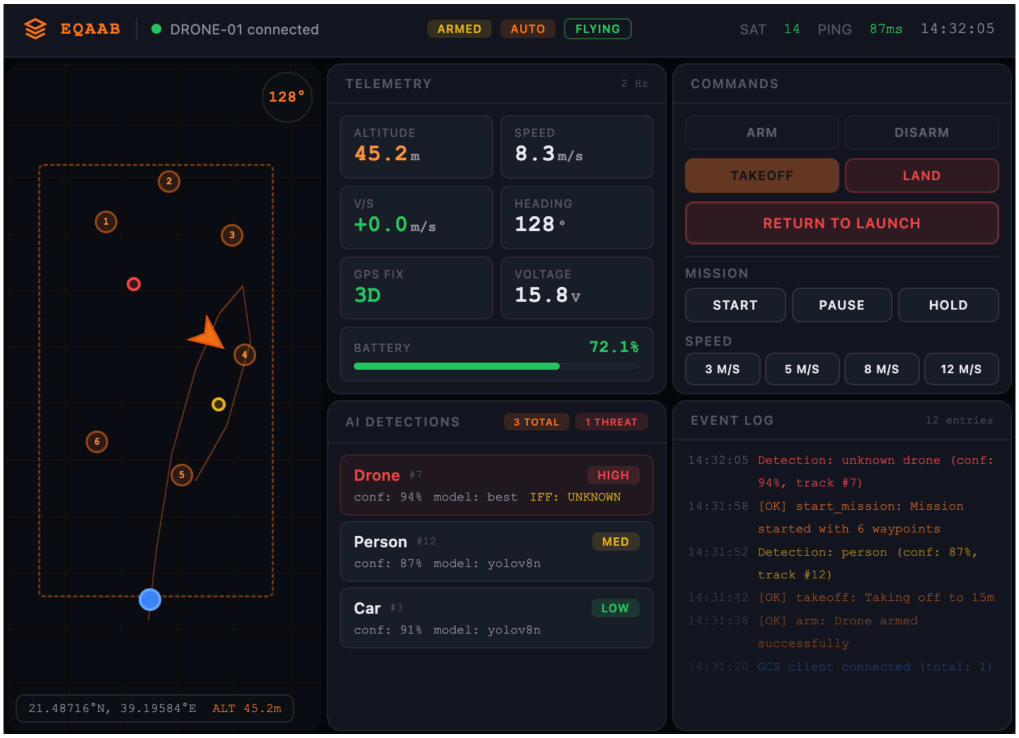
IFF System

The IFF (Identification Friend or Foe) node classifies every detected aerial drone target before any alert is generated. It queries an authorised registry — a JSON file containing the identifiers of all friendly aircraft operating within the surveillance zone. If the detected drone's identifier matches a registry entry, it is silently discarded. If no match is found, or if no identifier signal is received within the configured timeout, the target is immediately classified as Unknown and an alert is dispatched. This fail-safe default ensures that a malfunctioning IFF subsystem never suppresses a genuine threat. Person and vehicle targets bypass IFF evaluation entirely and are always forwarded as Unknown.



GCS Dashboard

The Eqaab Ground Control Station is a React.js single-page web application accessible through any standard browser without installation. It establishes a persistent WebSocket connection to the cloud backend on load and updates in real time as drone data arrives. The dashboard is built around four panels: a Leaflet.js interactive map displaying the drone's live position, flight path, and georeferenced detection alert markers; a telemetry status bar showing battery level, altitude, speed, and current flight mode; a chronological alert feed listing every detection event with its class, confidence score, and GPS coordinates; and a command panel allowing operators to issue high-level flight directives — Takeoff, Land, Return-to-Home, and waypoint upload. No video stream is displayed — all operationally relevant information is derived from structured detection data.



Results

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Conclusion

Eqaab demonstrates that a small, dedicated engineering team can design and deliver a complete intelligent aerial security platform — integrating autonomous flight control, real-time AI-based detection, IFF classification, cloud communication, and a live operator dashboard into a single cohesive system. The project validates the feasibility of fully onboard AI inference on resource-constrained embedded hardware, and establishes a concrete reference architecture for future UAV-based security deployments in border surveillance and critical infrastructure protection.

Future Work

- **Multi-drone coordination** — Extend the architecture to support simultaneous operation of multiple Eqaab units with a shared GCS interface and unified alert feed.
- **Thermal imaging** — Integrate an infrared camera module to enable reliable night-time and low-visibility detection beyond the capability of the current RGB pipeline.
- **Hardware-authenticated IFF** — Replace the current proximity-based identifier system with cryptographically signed transponder protocols for tamper-proof friend-or-foe classification.
- **Edge model optimisation** — Apply quantisation and pruning techniques to the YOLOv8 models to increase on-device FPS and extend mission duration on a single battery charge.